JOSEFFUS SANTOS

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EDUCATION

University of Manitoba — Winnipeg, MB Canada
Bachelor of Science Honours Computer Science DGPA 4.15 / 4.5 — Expected Dec 2024

University of Manitoba — Winnipeg, MB Canada
Bachelor of Science Major Biochemistry — Graduated May 2009

TECHNICAL SKILLS

- JavaScript
- Node.js
- Mocha
- CSS3 HTML
- Bootstrap
- Ubuntu
- Linux
- Windows
- Unix
- Python
- PyTest
- OpenCV
- Matplotlib
- Pandas
- Numpy
- Scikit
- Java
- JUnit
- Mockito
- Espresso
- Android Java
- C / C++
- Catch2
- C#
- SQL
- Cypher
- Git
- Qt
- VirtualBox
- Docker
- AWS
- GCP
- Azure
- SAS R
- NGINX
- fail2ban
- Fusion360
- Blender
- KiCAD
- Ghidra

RELEVANT EXPERIENCE

Research Technician — Feb 2011 - Present
Canadian Grain Commission, Government of Canada — Winnipeg, MB

- Developed programs in C# to extract data and add custom functions to instrument control software to speed up data analysis
- Developed VBA script used across Canada to automate data handling in Excel for the evaluation of research data
- Automated data analysis and generated charts with Python, R, SAS.
- AWS cloud services administration
  - Created a network of Windows and Linux EC2 instances to trial data analysis on the cloud
  - Created deployment scripts for users to automate ssh tunneling, instance start/stop, RDP connection
  - Set up Jupyter notebook servers accessible via browser over SOCKS5
- Wrote scripts to use AWS Textract to add searchable text to pdfs
- Designed and implemented robotic liquid sampling and dispensing systems using C++ and Arduino MKRzero / Atmel SAMD
- Taught other technicians and students on the basics of writing VBA scripts and Python for data analysis
- Communicated and resolved queries from clients external to the organization
- Maintained and troubleshot hardware and software of complex laboratory instrumentation
- Developed novel analysis methods for physical and chemical properties of grain, processing intermediates and derivatives.
- Contributed to data collection, analysis, and authoring research papers
- Created capital procurement tenders and evaluated technical specifications for complex laboratory equipment
- Planned research projects. Occasionally supervised teams of 4-15 students and technicians.

PROJECTS

Personal cloud / Home Lab | ZFS, Linux, GCP, AWS, Virtualbox, Docker, HTML, CSS, JavaScript — 2015 - Present

- Network of multiple LAN hosted, cloud hosted, and hypervisor hosted web services
- Mixture of LAN Linux and Windows hosts; Linux hosts on AWS that run applications in docker to provide VPN, Reverse Proxy, SOCKS5, SSH, remote desktop, content delivery, server monitoring and other services.

City Winnipeg Database | Python, Qt, SQLite, Google Maps API, GeoPandas, Pandas — 2021

- Combined multiple open data sources about the City of Winnipeg into a 3NF and BCNF normalized relational database
- Used Google Maps API to get asset geographic coordinates and used GeoPandas to measure distances and regional inclusions
- Created Windows, Linux cross platform GUI program in Python to run queries against SQLite DB

Meal Planner Calorie Tracking App | Android, Java, JUnit, Espresso, HSQLDB — 2022

- With a team of 5, designed and built a Java Android app following Agile development practices
- Applied software design patterns, implemented object oriented class design, and thorough unit testing
- Implemented a relational database in Hyper SQL DB to application data. Followed a 3 layer architecture.

Piston Pump | Fusion360, KiCAD, C++, Atmel SAMD — 2019 - 2022

- Designed and built a piston based liquid delivery device that exceeded ASTM class A standards for accuracy and delivery speed
- Designed touchscreen interface for human interaction, control, calibration, error recovery

PROJECTS cont

Drone AI person detection| C++, OpenCV, Python, OpenVino, MyriadX edge AI 2021 - Present
Created AI subsystem to autonomously identify people, measure distance, and track movement from an airborne drone

Mobile home controller | C/C++, HTML, CSS, JS, FreeRTOS 2022 - Present
Relay controller for mobile home appliances with cross platform embedded web page available via wifi AP on the controller

Fuzzy video content matcher | C/C++, Python, Ubuntu 2020 - 2022
Perceptual hashing to find similar videos. Implemented a python script on Linux to use Facebook TMK+PDQF algorithm

AR Pet | Android, Spark AR studio, Blender 2021
Android app prototype for an animated virtual pet visible via AR on a phone screen

3D Printer | Fusion360, C++, RAMPS, Arduino Mega, RaspberryPi 2014 - 2017
Designed and built an FDM 3D printer with control and monitoring via web interface

Led controller app | Android, Java, arduino 2017
App for selecting color and animation pattern for RGB led strips over bluetooth

Wireless controllable outlet hub | C/C++, HTML, CSS, JS, esp8266 2016
Lan/Wifi connected hub for controlling remote switches over 433mhz with cross platform embedded web page for control

Huffman encoder | Node.js, JavaScript, Mocha, OOP 2022
Huffman encoder implemented as a collection of JavaScript classes

Expert System | Prolog 2022
An expert system to solve the Farmer, Wolf, Goat, Cabbage problem

Poker with production system AI | Java, JUnit, OOP 2022
Standard poker with GUI where a Human player can play against a rule-based AI implemented as a collection of Java classes

Restaurant order queue simulator | C++, Catch2, OOP 2022
Event based order handling simulation, FIFO, LIFO and Priority Queue implemented as a collection of C++ classes

Maze pathfinder | Java 2022
Depth first and Breadth first search algorithms implemented in Java for maze pathfinding

Sudoku solver | Java 2021
Solve Sudoku puzzles of any size (within hardware limits).

Game of Life | C 2021
John Conway's Game of Life implemented in C

EXTRACURRICULAR

University Rocket club Avionics GUI Software | Git, C/C++, Qt, FreeRTOS, STMicro 2021 - 2022
Created cross platform desktop GUI application to allow avionics board configuration for general users

University Biomedical Engineering club Gaming Head Mouse | C/C++, Mbed OS 2021 - 2022
Created a head mounted gyroscopic joystick to allow a disabled gamer play with head movements

University Drone club website| HTML, CSS, JavaScript, Jekyll, Docker 2021
Contributed to the development of a club website

Keyloop Dealertech competition team proposal website| HTML, CSS, JavaScript, Github pages 2021
Created a website to show my teams proposed project. Won the university level competition

Magpie CTF Hackathon| Ghidra, IDA Free, Ollydbg, HTML, CSS, JS, C/C++, Go 2021
Participated in a capture the flag hackathon involving reverse engineering and penetration testing